ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PRACTICE EXAM 2 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Natural Sciences practice exam 2.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples. Penalise missing units once per question.

SECTION A

QUESTION 1

1 1.1 D ✓ 1.2 B ✓ 1.3 C ✓ 1.4 A ✓ 1.5 B ✓ 1.6 D ✓ 1.7 A ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ (10)

QUESTION 2

2 2.1 C ✓ 2.2 E ✓ 2.3 J ✓ 2.4 G ✓ 2.5 A ✓ 2.6 D ✓ 2.7 I ✓ 2.8 F ✓ 2.9 B ✓ 2.10 H ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1.1 To investigate how light intensity (distance of the lamp) affects the rate of photosynthesis ✓ (1)

3.1.2 The distance of the lamp from the plant (light intensity) ✓ (1)

3.1.3 The number of bubbles per minute (rate of photosynthesis) ✓ (1)

3.1.4 Oxygen ✓ (1)

3.1.5 The closer the lamp (the more light), the more bubbles were produced ✓, so the rate of photosynthesis increases as light intensity increases ✓. (2)

3.1.6 Any ONE: the same plant; the same volume and temperature of water; the same lamp; the same counting time ✓ (1)

3.2 glucose + oxygen ✓ → carbon dioxide + water ✓ + energy ✓ (3)

3.3 (a) The natural place where an organism lives ✓ (3)
(b) When organisms need the same limited resource (food, water, space, light, mates) ✓
(c) Any correct example, e.g. lions and leopards compete for impala ✓

3.4 At each step in a food chain most of the energy is used for life processes or lost as heat ✓ (only about 10% is passed on), so there is only enough energy to support a small number of carnivores ✓. (2)

3.5 (a) Bacteria, viruses, fungi, protists ✓✓ (one mark for any two) (5)
(b) Any TWO: wash hands with soap ✓; vaccination ✓; drink clean, boiled or treated water; cover your mouth when coughing; keep food covered and cooked properly; proper sanitation
(c) Any ONE: tuberculosis; cholera; typhoid; strep throat ✓

[20]

QUESTION 4

4.1 (a) The particles gain energy and vibrate faster ✓; they break free from their fixed positions and can slide past each other ✓. (4)
(b) Condensation ✓ (c) Sublimation ✓

4.2 The heat gives the air particles more energy, so they move faster ✓. They hit the inside of the balloon more often and harder ✓, so the pressure increases until the rubber can no longer stretch and it bursts ✓. (3)

4.3 (a) Honey at the bottom, then water, then oil at the top ✓✓ (b) On top of the oil ✓ (3)

4.4 (a) In the nucleus ✓ (b) Negative ✓ (c) The number of positive protons equals the number of negative electrons ✓ (3)

4.5 C ✓ Cl ✓ (2)

4.6 (a) magnesium + oxygen ✓ → magnesium oxide ✓ (4)
(b) Chemical change ✓; a new substance with different properties is formed (light and heat are given off) ✓

4.7 Any ONE: in a mixture the substances are not chemically joined and can be separated by physical methods; a compound has elements chemically joined in fixed amounts and can only be separated by a chemical reaction ✓ (1)

[20]

QUESTION 5

5.1 (a) Electrons are transferred between your shoes and the carpet, so your body builds up a static charge ✓; when you touch the metal (a conductor) the charge suddenly flows away, giving a small shock ✓. (3)
(b) A lightning conductor ✓

5.2 (a) $0{,}4\ \text{A}$ ✓ (4)
(b) $0{,}6 - 0{,}4$ ✓ $= 0{,}2\ \text{A}$ ✓
(c) In a parallel circuit, the total current divides between the branches (the branch currents add up to the total) ✓

5.3 (a) Any TWO: length ✓; thickness (cross-sectional area) ✓; type of material; temperature (3)
(b) Nichrome has a high resistance (and a high melting point), so it becomes very hot when current flows ✓

5.4 (a) Magnetic effect ✓ (b) Chemical effect ✓ (c) Heating effect ✓ (3)

5.5.1 Transparent lets almost all light through, e.g. clear glass ✓; translucent lets some light through but you cannot see clearly, e.g. frosted glass, tissue paper ✓; opaque lets no light through, e.g. wood, metal ✓ (3)

5.5.2 Sunlight enters raindrops and is refracted ✓; the different colours bend by different amounts (dispersion) and are reflected inside the drop, so the white light splits into a spectrum ✓. (2)

5.5.3 Black (dark) ✓; the green leaf reflects only green light and absorbs red, so no light is reflected to our eyes ✓. (2)

[20]

QUESTION 6

6.1.1 The further a planet is from the Sun ✓, the longer it takes to complete one orbit ✓. (2)

6.1.2 Mars ✓ (1)

6.1.3 $^{228}/_{150} \times 8$ ✓ ≈ 12 minutes ✓ (2)

6.1.4 It is very far from the Sun, so it receives very little heat ✓ (1)

6.2 Any TWO: it is a star ✓; it is made mostly of hydrogen and helium ✓; it produces energy by nuclear fusion; it is about 4,6 billion years old; it contains more than 99% of the mass of the solar system; it is about 150 million km from Earth (2)

6.3 Mercury ✓ and Venus ✓ (2)

6.4 (a) An optical telescope collects visible light ✓; a radio telescope collects radio waves ✓. (4)
(b) Above the atmosphere the images are not blurred by air, clouds or city lights ✓, and they can detect radiation that the atmosphere blocks ✓.

6.5 A group of stars that forms a pattern in the night sky ✓. Any ONE: Southern Cross (Crux), Orion, Scorpius, Centaurus ✓ (2)

6.6 A nebula ✓ (1)

6.7 The Moon turns once on its axis in the same time that it takes to orbit the Earth once ✓, so the same side always faces us ✓. (2)

6.8 The Large or Small Magellanic Cloud (accept the Andromeda Galaxy) ✓ (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE